A new 194-state machine for PA consistency

PA194H6, an audited comparison, and the remaining reduction problem

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Abstract

No machine with fewer than 194 working states is established in this report. The delivered fallback is PA194H6: a fully specified deterministic binary, single-tape machine, starting on a blank two-way infinite tape, with 194 working states and a separate halting label. It combines coefficient-six formula construction, no-push modus ponens, and a cache-only Shift test. The individual ideas occur in earlier packets, but this composition has a new literal machine graph and a different command behavior from its coefficient-six predecessor. The ordinary mathematical argument, reusing the explicitly identified numerical-calculus calibration, gives $\mathrm{PA} \vdash \neg\,\mathrm{Halt}(\mathrm{PA194H6}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})$. Two symbolic physical checks cover all 4,876 source outcomes, and the complete blank initialization is checked for 58,088 transitions. No new end-to-end Lean, Rocq, or iPRA theorem is claimed.

The audit pins ProveIt commit `3cc7fbf75ba1`. It covers the 47 filed research packets and all four incoming archives. A fresh literal-table census recognizes 2,049 complete numeric tables, with 683 byte identities; the 194-state frontier has 34 classes under the stronger comparison allowing state renaming, spanning roots, tape reflection, and global bit exchange. Reported constructions, auxiliary search inputs, and rejected machines are distinguished. PA194H6 matches none of those classes.

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1 Result, machine convention, and scope

The requested strict reduction was not obtained. PA194H6 meets the alternative of a materially different 194-state design. This is a composition result, not a claim that its three component ideas are individually new and not a claim that 194 is minimal.

The definitive table is `machine/machine.tm`. Its state set is $\{0, \ldots, 193\}$, its initial state is 0, and its terminal label is 194. The head initially scans coordinate 0 of an all-zero tape indexed by $\mathbb{Z}$. Each of its 388 transitions writes 0 or 1 and moves one cell left or right. The terminal label has no outgoing transitions and is excluded from the state count. Every working state is reachable in the directed transition graph from the start; graph reachability is not asserted to be blank-run reachability. Appendix A prints the whole table.

Quantity	Value
Working states / transition rows	194 / 388
Fixed controls / AX reader controls before folding	135 / 67
Simultaneous contextual folds / local cases	8 / 32
Primitive source sites / branch outcomes	3,775 / 4,876
Construction symbolic steps	1,567,458
Independent symbolic steps / longest case	1,139,739 / 403
Minimum positive physical-duration constant	3
Exact blank initialization	58,088 transitions

The SHA-256 of the delivered dense-table bytes is

`4bae5b0a698583d97ae8527472dfb627c1cecbba9babd88159fe94c1e3e7dc17`.

The native-table hash is

`9c1daeb54c6e2e402bc5a0405925c1137c19e26280bcf455efe4c91d9a61111d`.

State renaming from native start 74 and halt 254 is supplied in `machine/labels.json`. Whitespace and comments affect a file hash but not the machine comparison used below.

1.1 What is proved and what is inherited

The distinction between an ordinary mathematical proof and a checked proof-assistant artifact is substantive here. The pinned repository's PA README expressly says that its machine-specific iPRA reductions are not yet complete. Its imported PA196 development contains the PA implication from nonhalting to consistency, while later packets supply written arguments for both PA directions. This report uses those written proof transformations as identified mathematical dependencies; it does not promote their current formal status to a completed Lean or iPRA endpoint [1, 3].

Layer	Evidence and exact claim
Literal finite object	The entire table, start/halt labels, graph count, renaming and hashes are checked.
Local quotient	All 32 history/read cases are checked in the final simultaneous quotient, with their incoming-context hypotheses.

Layer	Evidence and exact claim
Source-to-tape execution	Two symbolic implementations check every primitive outcome. The independent implementation rederives the four register-list lemmas from literal rows and checks positive duration.
Source composition	Exact relocation of 61 multiplier bodies; all 3,409 other source sites and outgoing edges are checked. Restoring multiplier four gives the supplied PA194H graph exactly.
Command calculus	The displayed identities and proof-tree compiler are ordinary symbolic mathematics, accompanied by free-constructor checks.
Numerical calculus versus PA	The separated diagonal-head calibration and hereditary radix recoding are inherited ordinary mathematical proofs, retained in `references/logical/`. Their relevant finite implication proof template is also supplied.
Final theorem	Composition of those layers yields the stated ordinary PA-provability result. There is no newly compiled end-to-end proof-assistant certificate for that final theorem.

No consistency or reflection axiom is used to prove the equivalence. Assuming PA is consistent is needed for the further external conclusion that the machine in fact never halts.

2 Comparison of the supplied corpus

2.1 Snapshot and coverage

All claims about the supplied corpus refer to the pinned commit

`3cc7fbf75ba111a0da696ef2f6001dd528891bd5`.

The scope is `Computability/BusyBeaver/Research/PA/` and every ZIP then present in `docs/incoming/`. The 47 filed research packets and four incoming archives give a 51-packet ledger. The imported upstream snapshots and formalization directories are also included in the literal-table scan. The ledger in Appendix C and the CSV/JSON data identify the individual packages and their contributions.

The repository has already removed many repeated machine files, retaining mappings in `docs/OMITTED.tsv`. Its historical census describes 3,803 originally delivered table files and 481 classes, including unrooted table classes [2]. Our fresh scan instead measures tables actually available at this snapshot, together with tables inside the four incoming archives: 2,049 complete numeric tables and 683 exact byte identities. These counts are deliberately not equated. A script expecting one of the removed duplicate input paths may fail until that input is restored; that placement issue by itself says nothing about mathematical correctness. The table parser accepts complete five-column numeric transition tables in `.tm`, `.txt`, `.tsv`, and `.csv` files. Arbitrary JSON encodings are not included in these counts; packet-level source and proof contributions are assessed separately in the ledger.

Nested legacy archives were traversed through their actual maximum depth of eight; the 44 deepest occurrences add duplicate tables and no new byte identities.

The fresh 194-state frontier contains 208 occurrences, 62 byte forms and 34 classes under our stronger symmetry comparison. The filed packets contribute 30 such classes, and the incoming archives add PA194H, PA194HS/NS, PA194N5 and PA194N6. The comparison enumerates spanning graph roots when a start is not declared, normalizes states by deterministic traversal, and tests reflection and a

global symbol exchange as additional duplicate screens. A global symbol exchange is used only as a conservative novelty screen; it is not an assertion of blank-zero behavioral conjugacy.

These are inventories of encoded objects, not 34 proved consistency machines. Twenty classes have supplied ordinary PA arguments and independent symbolic physical evidence; N5 has an ordinary argument and one universal JavaScript symbolic audit. Nine are auxiliary research inputs, and four are explicitly rejected. The full 34-row frontier states that status for each class.

2.2 Main design families

Family	States	Main distinction and implication
Unary / U208 / PA209	208 and larger	Earlier proof-search and physical interpreter presentations. Several U208 packets repeat the same main table. A physical fold cannot be transferred by its old state numbers after the source and allocation have changed.
PA196	196	James Chang's compiled numerical PA-refutation search supplies the central source, interpreter, and logical proof-producing foundation.
Diagonal / separated heads	199, 198, 196, 195, 194	Diagonal vacuity $B \to \forall x_B B$ replaces a larger equality-vacuity component. Separating that head from the gate changes the stack interface from nine to ten pushed entries. A smaller source does not automatically mean a smaller compiled machine.
Primary PA194 / 68–8	194	Same separated-head logic, but different decoder DAGs: 67 readers minus seven folds, or 68 readers minus eight folds. These are distinct literal machines. The AD alternative exchanges physical register allocation.
C / CS	194	Cached MP result and cache-only Shift, with CS also using a symmetric comparison. Five headline packages repeat the primary C class.
CSZ / J / ZR / A / F / R	194	Replace suitable jumps by decrements of a proved-zero register, or reassociate the decoder. These are controller or allocation changes, not new consistency strengths. CSZ and X are the same dense table.
Q / D	194	Q exchanges curried premises in a quantifier head; D emits $\Gamma(D)$ instead of $\Gamma(A)$. Their proof selectors must change with the emitter.
M4R / M5 / M6	194	Reassociated multiplier-four decoder or multipliers five and six. The latter require hereditary recoding of both formulas and variable labels and a treatment of unused residues.
N / N5 / N6	194	MP returns its conclusion in the visible top without a dummy-zero push. N5 and N6 combine this rule with the corresponding formula multipliers.
H / HS = NS	194	No-push MP plus cache-only Shift at multiplier four. HS uses the symmetric implementation. HS and NS are byte-identical.

Family	States	Main distinction and implication
H6, this report	**194**	Multiplier six, no-push MP, and cache-only Shift. New combined controller/encoding semantics and new literal graph; no state saving is claimed.

The two incoming presentations of N6 are byte-identical as well. The compact N5/N6 archive and the larger N6 research archive therefore contribute one N6 machine between them. These identifications matter when deciding whether a proposed fallback is actually new.

2.3 Why PA194H6 is materially different

The changes operate during the repeated proof search. They are not confined to initialization, unreachable rows, state names, tape reflection, or a rewriting of the final halting action. Relative to N6, the Shift test changes from $t + h = 1$ to $h = 1$. Thus N6 halts on Shift from $(t, w, d, h) = (1, 0, 0, 0)$, whereas H6 clears that store without halting. A proof of contradiction in the visible top must therefore be exposed by Pop, not by a single final Shift.

Relative to the multiplier-four hybrid H, the formula constructor and MP comparison change. Since $p(0, 2) = 5$, H accepts the MP comparison at $(0, w, 0, 20)$ with parameter 2, while H6 rejects it; H6 instead accepts cache 30. The Cantor stack and word scheduler still use unscaled pairing. Therefore changing formula codes is not a relabeling of complete stores or an assertion that their arithmetic execution traces are conjugate.

These examples distinguish specified command functions on natural stores; we do not use them to claim that the same stores occur on the actual blank run under a consistency assumption. Independent full-graph comparison establishes the literal novelty against all 34 old frontier classes.

3 The numerical calculus and its PA calibration

This section specifies the logical system searched by H6 and summarizes the ordinary proof dependency. The complete inherited calibration texts are preserved, including the diagonal replacement, the separate-head interface, the radix proof and the original PA195 calibration.

Let $p(a, b) = (a + b)(a + b + 1)/2 + b$ and define

$$I(a, b) = 6p(a, b), \quad M(a, b) = 6p(a, b) + 1, \quad U(i, a) = 6p(a, i) + 2, \quad E(a, b) = 6p(a, b) + 3.$$

The numerical formulas are all natural numbers, with code 0 interpreted as truth and code 1 as the designated contradiction. Write $A \to B = I(A, B)$, $\neg A = I(A, 1)$, $A \wedge B = \neg(A \to \neg B)$, and $\exists_i A = \neg U(i, \neg A)$. Codes with residue four or five are allowed; the soundness interpretation maps them to truth.

The six universal axiom families are

$$\begin{aligned}
L_1(D, H) &= (\exists_D U(D, H)) \to H, \\
L_2(D, A, H) &= U(D, U(A, H)) \to U(A, U(D, H)), \\
L_3(D, A, H) &= M(H, D) \to (E(H, A) \to M(A, D)), \\
L_4(D, A, H) &= E(H, D) \to (M(A, H) \to M(A, D)), \\
L_5(D, H) &= \exists_D E(H, D), \\
L_6(D, A, H) &= U(D, A) \to (U(D, A \to H) \to U(D, H)).
\end{aligned}$$

Add the dominated implicational family

$$\rho_8(D, A, H, u) = ((H \to u) \to D) \to ((D \to H) \to (A \to H)), \qquad u \le D,$$

and every diagonal-vacuity formula $\Gamma(B) = B \to U(B, B)$. For the final axiom family put

$$\begin{aligned} V(a,b,c) &= M(a,b) \to U(c, M(a,b)), & P(a,b,c) &= c \to M(c,b), \\ S(a,b,c) &= U\big(b, (M(a,b) \to P(a,b,c)) \wedge (M(c,b) \to c)\big), \\ Q(a,b,c) &= U(c, S(a,b,c) \to \neg P(a,b,c)), \\ G(a,b,c) &= \neg(V(a,b,c) \to Q(a,b,c)). \end{aligned}$$

The gate $G(a, b, c)$ is an axiom when $c \neq a$ and $c \neq b$. The only inference rules are modus ponens and theorem generalization $B \mapsto U(i, B)$. There are no undischarged hypotheses in this numerical proof system. Code zero has a fixed derivation from $\rho_8(0, 0, 0, 0) = 0$.

3.1 Why the calibration survives multiplier six

The dependency is a syntactic, PA-verifiable pair of refutation translations, not merely a standard-model equiconsistency assertion. Four points prevent incorrect shortcuts.

First, formula codes also serve as variable labels in $\Gamma(B)$. The hereditary recoding F from coefficient four to coefficient six is

$$F(0) = 0, \qquad F(4p(a,b) + r) = 6p(F(a), F(b)) + r \quad (n > 0,\ 0 \le r < 4).$$

Every recursive argument is smaller than its input, so course-of-values primitive recursion defines F. A similarly recursive partial inverse proves injectivity. It follows that constructors, binding, repeated labels, guards and diagonal heads are preserved. Keeping labels fixed while recoding formulas would fail to preserve the diagonal head.

Second, F need not preserve the side condition $u \le D$. For example, $40 < 44$, but $F(40) = 126 > 66 = F(44)$. The retained finite `RestrictedImplication.lean` proof template derives unrestricted ρ_8 from the dominated schema using only reflexivity and closure of domination under either implication argument. At coefficient six, these follow from $a, b \le p(a, b) \le 6p(a, b)$. Translate each old dominated leaf to that finite unrestricted derivation, whose new restricted leaves satisfy their own target inequalities. No monotonicity of F is assumed.

Third, the translation back into PA interprets membership by the arithmetic bit predicate, equality by equality, the quantifier by ordinary universal quantification, and residues four and five by truth. Every label in a decoded formula of nonzero code B is less than B, so the diagonal head has a uniform vacuity proof. The numerical alias $I(0, 0) = 0$ is handled by a proof of $(0 = 0) \leftrightarrow ((0 = 0) \to (0 = 0))$, rather than a false assertion of syntactic identity.

Fourth, the guarded gate has a uniform PA proof supplier using finite bit-set comprehension. For each arithmetic formula θ, PA proves

$$\forall n\ \exists s < 2^n\ \forall z < n\ (\mathrm{Bit}(s, z) \leftrightarrow \theta(z)).$$

A supplier inserts the appropriate induction instance and excluded-middle case split. It does not decide θ and does not use a truth or reflection principle. The separate case $a = b$ in the gate uses the irreflexivity $\neg\,\mathrm{Bit}(z, z)$, and the guard makes the fresh label usable. These cases and the full matrices are written out in the retained radix calibration.

The coefficient-four completeness construction is reused through its three required equality interfaces: two-way membership rewriting, neighborhood equality, and an equality witness. The diagonal replacement

recovers those interfaces with explicit proof templates, using the code of a negated target formula as its fresh label. The independent head Γ and the gate are interderivable with the earlier combined gate by classical implication proofs. This identifies precisely where the earlier proof-producing PA compiler is used [3, 4, 5].

Thus there are primitive-recursive proof transformations α, β whose correctness is provable in PA:

$$\mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner) \Longrightarrow \mathrm{Prf}_{\mathcal{L}_6}(\alpha(p), 1), \qquad \mathrm{Prf}_{\mathcal{L}_6}(q, 1) \Longrightarrow \mathrm{Prf}_{\mathrm{PA}}(\beta(q), \ulcorner 0 = 1 \urcorner).$$

These are the logical inputs to the following machine-specific proof.

4 PA194H6: the coefficient-six cached-test controller

PA194H6 combines coefficient-six formula constructors with the no-push modus-ponens rule and a Shift command that tests only the cached formula. Coefficient six and no-push modus ponens occur together in the supplied PA194N6. The cached-test/no-push interface occurs at coefficient four in PA194H and PA194HS/NS. The contribution here is their verified composition, with a new literal graph at 194 states. The component ideas are inherited. The composition does not lower the state count.

4.1 Exact source and physical construction

The native table starts at 74 and halts at the separate label 254. Relative to PA194N6, precisely the following five rows change:

State	Read	Old action	New action
38	0	$0R28$	$0R11$
38	1	$1R20$	$1R122$
214	0	$0R188$	$0R312$
214	1	$1R312$	$1R20$
312	0	$0R11$	$0R188$

The no-push success edge $(164, 1) \mapsto (1, R, 114)$ is retained. The 135 fixed controls and 67 AX readers form a 202-state parent. Eight simultaneous history-sensitive folds give $135 + 67 - 8 = 194$ states. All 32 history/read-symbol cases are rechecked in the final quotient; all arithmetic-box transitions are unchanged from the coefficient-six parent.

The source edits relative to the coefficient-six push-MP parent are

$$\texttt{node163} = (\texttt{node6}, \texttt{node54}),$$
$$\texttt{node170} = ((\texttt{node36}, (\texttt{node167}, \texttt{node145})), \texttt{node57}).$$

The first removes the successful-MP dummy-zero push. The second implements cached-only Shift: it clears the old top, copies the cached value h into T and X, restores X to H, sets $X = 1$, and enters the inherited comparison. Thus the comparison receives $(T, H, X) = (h, h, 1)$. It halts precisely when $h = 1$; otherwise it clears its scratch operands and continues to the unchanged Cantor-stack decoder.

There is an independent exact source check. Replacing the coefficient-six multiplier subtree by its four-increment predecessor gives exactly the supplied PA194H instruction graph, not merely the same number of sites. The new graph replaces 61 four-increment blocks by six-increment blocks. Every one of the other 3,409 sites and every outgoing edge agrees under the explicit relocation, and no outside edge enters an interior point of a replacement block. Consequently 3,653 sites become 3,775 sites.

The generic constructor proof retained in the references establishes that these replacements implement $6p(a,b)$ instead of $4p(a,b)$ throughout AX and the MP comparison. They leave the Cantor stack pairing unchanged.

The construction checker proves all 4,876 primitive outcome contracts. A separately implemented literal-tape checker rederives four register-list lemmas from the transition rows and proves those same contracts over arbitrary natural registers, both decrement outcomes, an arbitrary affine header gap, and any nonempty finite list of natural spare values. Its 1,139,739 symbolic operations have maximum case length 403 and minimum positive physical-duration constant three. Complete blank initialization takes exactly 58,088 literal TM steps and reaches source entry 25 with logical registers $(2,0,0,0,0,0,0,0,0)$ and 26 unit spare blocks. The entire finite-support tape, head, control, and zero exterior are checked. These are executable universal certificates, not proof-assistant output.

4.2 Numerical semantics and inherited logical calibration

Put

$$p(a,b)=\frac{(a+b)(a+b+1)}{2}+b,\qquad [a;w]=p(a,w),\qquad I(a,b)=6p(a,b),\qquad U(i,a)=6p(a,i)+2.$$

Membership and equality codes are $M(a,b)=6p(a,b)+1$ and $E(a,b)=6p(a,b)+3$. Negation is $N(a)=I(a,1)$; conjunction and existential quantification have the inherited definitions $C(a,b)=N(I(a,N(b)))$ and $\exists_i a=N(U(i,N(a)))$. The codes zero and one are truth and the target contradiction. The unused residue classes four and five are interpreted by the inherited radix calibration; they are not silently assumed to be ordinary formula constructors. In particular $I(0,0)=0$ remains a numerical alias.

Let $\mathcal{L}_6$ denote the supplied separated-head numerical calculus with these coefficient-six constructors. The logical dependency used here is the pair of primitive-recursive proof translations α,β whose correctness is provable in PA:

$$\begin{aligned}\mathrm{Prf}_{\mathrm{PA}}(p,\ulcorner 0=1\urcorner)&\longrightarrow \mathrm{Prf}_{\mathcal{L}_6}(\alpha(p),1),\\ \mathrm{Prf}_{\mathcal{L}_6}(q,1)&\longrightarrow \mathrm{Prf}_{\mathrm{PA}}(\beta(q),\ulcorner 0=1\urcorner).\end{aligned}$$

The retained `radix_calibration.tex`, `separate_head_calculus.tex`, and original calibration give this ordinary mathematical interface. The retained `RestrictedImplication.lean` supplies the explicit unrestricted ρ_8 proof template (`rho8All`) needed by the recoding argument; this avoids assuming that recoding preserves the numerical dominance order. This release does not supply a newly compiled end-to-end Lean or Rocq proof of it, nor an iPRA endpoint. Nothing in the new argument assumes PA's consistency or its own soundness.

A command store is (t,w,d,h), with current top, stack, numerical offset, and cached formula. For $b=d+x$ the two controller operations are

$$S(t,w,d,h)=\begin{cases}\mathsf{halt}, & h=1,\\ (h,\mathrm{tl}\,w,d,\mathrm{hd}\,w), & h\neq 1,\end{cases}\qquad \mathsf{MP}_x(t,w,d,h)=\begin{cases}(b,w,b,0), & I(t,b)=h,\\ (0,w,b,0), & I(t,b)\neq h.\end{cases}$$

The total Cantor projections obey $\mathrm{hd}[a;w]=a$, $\mathrm{tl}[a;w]=w$, and $\mathrm{hd}\,0=\mathrm{tl}\,0=0$. Thus $[0;0]=0$; no proof below treats the numerical stack as a free list.

For $D=d+x$, $H=h+y$, and $A=c$, the unchanged AX operation returns

$$\mathsf{AX}_{c,x,y}(t,w,d,h)=(G(H,D,A),[q_1,\dots,q_9,t;w],0,0).$$

Its first head is $q_1=I(A,U(A,A))$, its next six heads are the inherited universal heads in reverse emission order, and

$$q_8=I(I(I(H,d),D),I(I(D,H),I(A,H))),\qquad q_9=U(d,h).$$

The guarded gate G is exactly the coefficient-six parent's gate. The complete free-constructor emitter is checked in `proof/check_commands.py`. All $q_1, \dots, q_8, G$ are zero modulo six, whereas q_9 is two modulo six and at least two. None is one. This constructor observation is valid for arbitrary input naturals and requires no semantic consistency assumption.

4.3 A complete command compiler

Write $A_0 = \mathsf{AX}_{0,0,0}$ and execute words left to right. The exact identities

$$\mathsf{AX}_{c,x,y}; S^9(t,w,d,h) = (q_8, [t;w], 0, U(d,h)), \qquad \mathsf{AX}_{c,x,y}; S^{10}(t,w,d,h) = (U(d,h), w, 0, t)$$

never halt. Their tests encounter generated non-one heads and the cleared cache, not the arbitrary saved top. Define

$$\mathsf{Push} = A_0; S^9; \mathsf{MP}_0, \quad \mathsf{Pop} = A_0; S^{11}, \quad \mathsf{Discard} = \mathsf{MP}_0, \quad \mathsf{Offset}_i = \mathsf{MP}_i.$$

They satisfy, for every displayed natural store,

$$\begin{aligned} \mathsf{Push}(t,w,d,h) &= (0, [t;w], 0, 0), \\ \mathsf{Pop}(t, [a;w], d, h) &= \begin{cases} \mathsf{halt}, & t = 1, \\ (t, w, 0, a), & t \neq 1, \end{cases} \\ \mathsf{Discard}(t,w,0,h) &= (0,w,0,0), \\ \mathsf{Offset}_i(0,w,0,0) &= (0,w,i,0). \end{aligned}$$

Push's final comparison fails by the residues zero and two. Pop's eleventh Shift tests precisely the saved top. The two MP branches coincide when the candidate conclusion is zero, proving Discard even with arbitrary old cache. Offset includes the zero case. All identities continue to hold when a zero push collapses into the empty stack.

From $(t,w,0,0)$, the word $\mathsf{Discard}; \mathsf{AX}_{A,D,H}; S^{j+1}; \mathsf{Pop}^{9-j}$ selects q_j for $1 \le j \le 7$, preserving w and clearing both caches. The gate uses $\mathsf{Discard}; \mathsf{AX}_{c,b,a}; \mathsf{Pop}^{10}$. For $u \le a$, the dominated implicational head is selected by

$$\mathsf{Discard}; \mathsf{Offset}_u; \mathsf{AX}_{b,a-u,c}; S^9; \mathsf{Pop}.$$

Generalization is

$$\mathsf{Gen}_i = \mathsf{Push}; \mathsf{Offset}_i; S; \mathsf{AX}_{i,0,0}; S^{10},$$

which sends $(a,w,0,0)$ to $(U(i,a),w,0,0)$, including $a = 1$. These are the inherited hybrid words; their proof uses only the constructor properties just verified for coefficient six.

The MP suffix $S; S; \mathsf{MP}_b; \mathsf{Pop}$ sends $(0, [a, I(a,b), 0; w], 0, 0)$ to $(b, w, 0, 0)$ when $a, b \neq 1$, and halts when $a = 1$ or $b = 1$. Indeed the two Shifts expose the premises; successful MP leaves $(b, [0;w], b, 0)$; Pop consumes the reserved zero and clears the offset, or detects the false conclusion. The reserved zero must be supplied by the proof compiler, since MP no longer pushes it.

For a finite proof tree p, compile axioms by these selectors and a universal-introduction node by appending Gen_i. At an MP node with subproofs p_0 of a and p_1 of $I(a,b)$, use

$$W_p = \mathsf{Discard}; \mathsf{Push}; W_{p_1}; \mathsf{Push}; W_{p_0}; \mathsf{Push}; S; S; \mathsf{MP}_b; \mathsf{Pop}.$$

Induction on p proves that execution from every $(t,w,0,0)$ either halts or returns $(\mathrm{concl}\, p, w, 0, 0)$. The first two commands reserve the zero needed by the suffix. If p proves one, the word $A_0; W_p; \mathsf{Pop}$ therefore halts from every incoming store. The initial AX clears arbitrary carried offsets and cache; the final Pop tests one before using the old underlying stack. A plain final Shift would be incorrect: here $S(1,0,0,0) = (0,0,0,0)$.

4.4 Exhaustiveness, soundness, and the PA theorem

The unchanged scheduler tests finite words while carrying the store between trials. The preceding all-store compiler is consequently essential. Its continuation convention is $E(\epsilon) = 0$ and $E(c :: w) = 1 + B_c(E(w))$, where

$$B_S(z) = p(0, z), \quad B_{\mathsf{MP}_x}(z) = p(1, p(x, z)), \quad B_{\mathsf{AX}_{c,x,y}}(z) = p(c + 2, p(y, p(x, z))).$$

Let $z = E(A_0; W_p; \mathsf{Pop})$. Prefixing the halting word by $\mathsf{AX}_{k,0,0}$ gives the fresh-candidate codes $n_k = p(k + 2, p(0, p(0, z)))$. They are increasing and unbounded. Each prefix may alter the store arbitrarily because the following word halts from every store. The fresh counter begins at two; a member of this cofinal family will be reached unless an earlier halt occurs. Positive continuations strictly decrease during decoding. The inherited source arithmetic loops terminate with primitive-recursive bounds, so each earlier finite trial also terminates or halts. These are finite coding inductions formalizable in PA, not an appeal to a consistent standard model.

For soundness, annotate the current top, cache and each canonical Cantor stack entry by a derivation in $\mathcal{L}_6$. Zero has a fixed derivation and supplies the virtual zero tail. AX preserves the annotations by its axioms and generalization; successful MP uses its two annotated premises; failed MP returns zero; Shift moves annotations and halts only when an annotated cache equals one. The canonical stack has a finite bounded decoding because its tail decreases at each positive value. Proof annotations can therefore be updated by primitive recursion on a finite execution. PA proves the soundness invariant and the proof-tree compiler by ordinary induction.

Composing these two syntactic constructions with α, β, the exact source binding, the positive universal physical simulations, and the complete blank initialization gives the ordinary mathematical theorem

$$\mathrm{PA} \ \vdash \ \mathrm{Halt}(\mathrm{PA194H6}, \mathrm{blank}) \ \longleftrightarrow \ \exists p \ \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner),$$

and hence

$$\mathrm{PA} \ \vdash \ \neg \, \mathrm{Halt}(\mathrm{PA194H6}, \mathrm{blank}) \ \longleftrightarrow \ \mathrm{Con}(\mathrm{PA}).$$

The logical calibration is an explicitly inherited ordinary-proof dependency. The new physical and symbolic checks support the composition and do not constitute an end-to-end proof-assistant theorem. Consistency is needed only for the further external conclusion that the machine actually never halts.

5 Reduction attempts and their limits

The comparison prevents three misleading ways of claiming a new reduction. One cannot add savings from folds that identify overlapping source roles; one cannot subtract a control needed by blank initialization just because it is absent from a later invariant; and one cannot count a shorter source word as a state saving before constructing and checking its actual decoder. Several rejected tables in the supplied corpus illustrate these issues.

5.1 The six-zero scanner obstruction

The new candidate still uses a scanner recognizing a run of seven zeros. Deleting native state 34, redirecting its incoming target to 75, would leave 193 working states. This straightforward proposal fails the old scanner contract even at a context reached by the parent from blank tape.

After the checked 58,088-step startup, the parent is at source site 25, whose path is `1000000`; the instruction is decrement of logical register 8, physically register 1, which is zero in that seed. After another 56 parent transitions, the left scan enters native state 2 at coordinate 44. Starting from that exact complete tape:

Scanner	Steps to its exit	Exit state	Exit head
Seven-zero parent	64	84	−2
Six-zero deletion	47	84	−1

The two exit tapes are identical; the head position is wrong by one cell. The longer source test returns to the intended next canonical frame after 438 parent steps. The local finite witness already suffices to reject the claim that this deletion preserves the specified scanner operation. It is not an assertion about the complete altered machine's eventual halting. The standalone script `research/scanner_probe.py` reproduces every cell and step of the witness.

The path `1000000` is in the fixed scheduler scaffold outside AX. Consequently changing only the AX grammar or its head order cannot make the unchanged six-zero scanner contract valid. A successful scanner reduction must also change that scaffold, its addressing convention, or the scanner implementation. This is a limitation of that proposal, not a lower bound for all 193-state machines.

5.2 Additional decoder and contraction searches

Fresh bounded searches combined seven-head permutation and bijective parameter renaming with continuing shared and occurrence-level binary-tree reassociation. Earlier packets already considered seven-head permutations in fixed shapes; the added scope is that combination with reassociation and several controller parents. For coefficient six, the search also recognized the existing three-increment reader at native state 330, which was absent from the earlier grammar-interning map. It allowed eligible contextual folds involving that fixed reader. The recorded best effective counts remained compatible with 194 states; no verified 193-state table resulted. The exact proposal counts, pseudorandom seeds, objective values, bounds and search inputs are supplied in `research/decoder/`. The five completed runs used 36,567,758 proposal slots. These are not counts of distinct machines. All 301 saved grammar words were independently checked to belong to the stated seven-head mutation domain; that check does not certify their physical execution.

Run	Proposal slots	Best state-count score
Primary controller	10,827,557	194
Cached controller	10,700,438	194
More frequent permutations	5,621,901	194
H6, triple-increment reuse	5,590,551	194
H6, diagonal-D initial bias	3,827,311	194

A whole-source-contract contraction screen on H6 considered 62,073 specified row-choice merges while protecting the arithmetic block. It found no survivor at its stated symbolic fuel bound. This rules out those tested contractions under that proof interface, not contractions requiring a changed source invariant or a different interpreter. Two further finite local screens used eleven-cell windows and at most 300 local transitions: arbitrary common replacement actions processed all 194 controls, and independent incoming-edge bypasses processed 172 eligible controls. Neither returned a contraction. The saved code and receipts define the search space more precisely than a claim that the machine is irreducible.

There is a further non-additivity obstacle. The supplied symmetric hybrid uses a new reader at native label 330 and a fold consuming state 41. The coefficient-six constructor already uses 330 for three increments and its allocation also consumes state 41. Renaming the conflicting reader to 331 produces a source-correct 195-state symmetric H6 implementation. Blindly adding the two advertised savings double-counts the same available control. The accepted H6 uses the ordinary cached-only comparison, with its separately verified eightfold allocation.

5.3 Ideas from the ZFC and ConciseIPRA work

The useful transferable ideas are semantic interfaces and verified compilation: remove an unnecessary stack push, use a known-zero register to implement a jump, separate a short logical basis from its search procedure, and exploit context when allocating decoder states. The inspected ZFC work already uses related no-push and context-fold techniques; the present report does not claim those in isolation as new [8].

The numerical fresh-label trick and separated diagonal head are inherited from the PA196/PA194 line. ConciseIPRA supports the broader strategy of replacing a large presentation by a smaller derivably equivalent interface, but its target is iPRA. Merely searching for an iPRA refutation would not establish the requested PA-consistency equivalence. A transfer needs explicit PA-refutation translations and their compiled cost. Likewise, substituting a ZFC-consistency search changes the target strength [9].

6 Reproduction and package structure

The accepted construction uses Python 3.10 or later and the standard library. No external checkout, network connection, proof assistant, or heuristic search is needed for the package's default verification command. Run with assertions enabled:

```
python3 -B verify.py
```

The verifier rebuilds the table in a temporary directory, checks its expected bytes and source graph, checks local folds, runs both symbolic physical implementations, checks the complete startup, and verifies the new command and source-change identities. The separate comparison verifier checks novelty against the bundled old 194-state tables. The optional discovery searches are not a premise of the accepted mathematical result.

To rebuild the report from its source directory:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The package retains licenses and source provenance for the inherited compiler, symbolic engines and proof fragments. `MANIFEST.json` records the SHA-256 of every delivered file other than itself. The report's LaTeX source, compiled PDF, complete table, verifiers, comparison ledger and relevant research evidence are all contained in the ZIP. Historical claims about earlier packages are identified as such; their entire verification suites were not all rerun.

A Complete 194-state transition table

An entry wDn means write w, move in direction D, and enter state n. H denotes terminal label 194. The start is state 0 and the blank symbol is 0. All 194 rows below have both read-symbol actions.

State	Read 0	Read 1	State	Read 0	Read 1
0	$1R1$	$1R2$	97	$0R36$	$1R114$
1	$1R3$	$1R1$	98	$0R113$	$1R75$
2	$0R0$	$1R2$	99	$0L63$	$0R115$

State	Read 0	Read 1	State	Read 0	Read 1
3	$0R4$	$0R1$	100	$0L62$	$1R116$
4	$1R5$	$1L6$	101	$0L90$	$1R117$
5	$1L7$	$1L8$	102	$0R118$	$0L14$
6	$0L7$	$1L6$	103	$0R69$	$1R103$
7	$0L9$	$1L6$	104	$0R119$	$1R95$
8	$0L10$	$0L11$	105	$0R119$	$1R67$
9	$1R12$	$0L9$	106	$0R120$	$0L14$
10	$0L9$	$0L10$	107	$0R121$	$1L71$
11	$1L8$	$1L11$	108	$0R122$	$1R123$
12	$0R12$	$0R13$	109	$0R112$	$1R109$
13	$0L14$	$0R13$	110	$0L124$	$1R125$
14	$1L15$	$0L15$	111	$0R109$	$1R111$
15	$0L15$	$1L16$	112	$0R115$	$1R112$
16	$0L17$	$1L16$	113	$0R126$	$1L35$
17	$0L18$	$1L16$	114	$0R127$	$1R128$
18	$0L19$	$1L16$	115	$0R14$	$1R115$
19	$0L20$	$1L16$	116	$0R88$	$0L14$
20	$0L21$	$1L16$	117	$0R90$	$1L95$
21	$0L22$	$1L16$	118	$0R129$	$1R129$
22	$0L23$	$1L16$	119	$0L63$	$0R103$
23	$0R23$	$1R24$	120	$0R68$	$0R91$
24	$0R25$	$1R26$	121	$0R58$	$0R91$
25	$0R27$	$1R28$	122	$0L86$	$1R130$
26	$0R29$	$1R30$	123	$0R131$	$1R132$
27	$0R31$	$1RH$	124	$0R133$	$1R134$
28	$0L32$	$1R33$	125	$0R135$	$1R136$
29	$0R34$	$1R35$	126	$0R58$	$0R111$
30	$0L36$	$1L37$	127	$0R99$	$1R116$
31	$0R38$	$1R39$	128	$0R79$	$1R78$
32	$0R40$	$1R41$	129	$0R88$	$1R47$
33	$0R42$	$1R43$	130	$0R137$	$1R138$
34	$0R44$	$1R45$	131	$0R32$	$1R36$
35	$0R46$	$1R33$	132	$0R31$	$1R98$
36	$0R47$	$1R48$	133	$0R139$	$1R88$
37	$0R49$	$1R50$	134	$0L138$	$1R140$
38	$0R51$	$1R42$	135	$0R41$	$1L141$
39	$0R52$	$1R53$	136	$0R142$	$1R143$
40	$0R54$	$1R55$	137	$0R136$	$1R144$
41	$0R56$	$1R57$	138	$0R145$	$1R146$
42	$0R58$	$0R59$	139	$0R147$	$1R148$
43	$0R60$	$0L14$	140	$0R143$	$1R136$
44	$0R61$	$1R62$	141	$0R149$	$1R143$
45	$0R63$	$1R52$	142	$0R97$	$1R35$
46	$0R64$	$1R65$	143	$0R150$	$1R47$
47	$0L46$	$1L66$	144	$0R151$	$1R152$
48	$0R67$	$0L14$	145	$0R153$	$1R66$
49	$0R68$	$0R69$	146	$0R154$	$1R155$
50	$0R70$	$0L48$	147	$0R97$	$1R156$

State	Read 0	Read 1	State	Read 0	Read 1
51	0R64	1R53	148	0L136	1R88
52	0L71	0L15	149	0R148	1R41
53	0R72	0L10	150	0R76	1R41
54	0R73	1R74	151	0R157	1R158
55	0R75	1R76	152	0R134	1R159
56	0R77	1R78	153	0R142	1R160
57	0R79	1R80	154	0R147	1L144
58	0L81	0R1	155	0R161	1R162
59	0R82	1R59	156	0R107	1R85
60	0L73	1L46	157	0R28	1R163
61	0R83	1R84	158	0L164	1L165
62	0R54	1R85	159	0R166	1R167
63	0L86	1R87	160	0R147	1R135
64	0R58	0R82	161	0R168	1R169
65	0R88	1R30	162	0R170	0L10
66	0R88	1R89	163	0R171	1R172
67	0L90	1L33	164	0R173	1R167
68	0L81	0R2	165	0R174	1R166
69	0R91	1R69	166	0R160	1R41
70	0R72	1R92	167	0R142	1R148
71	0R46	1R93	168	0R175	1R124
72	0R42	0L13	169	0R176	1R177
73	0R94	1R37	170	0R178	1R179
74	0R30	1R95	171	0R180	1R181
75	0R42	1R96	172	0L182	1L183
76	0R97	1R98	173	0R147	1R154
77	0R99	1L100	174	0R172	1R184
78	0R30	1R77	175	0R66	1R172
79	0R93	1R101	176	0R135	1R185
80	0R30	1R102	177	0R186	1R107
81	1R81	1L15	178	0R31	1R52
82	0R103	1R82	179	0R187	1R178
83	0R94	0L7	180	0R142	1R150
84	0R104	1R105	181	0R32	1R188
85	0R42	1R106	182	0R158	1R189
86	0R107	1R48	183	0R190	1R191
87	0R108	0L72	184	0R41	1R192
88	0R68	0R109	185	0R164	1R159
89	0R110	1R88	186	0R172	1R183
90	0R49	1R100	187	0R92	1R113
91	0R111	1R91	188	0R193	1R110
92	0R30	1R48	189	0R141	1R165
93	0R47	1R60	190	0R135	1R139
94	0L63	0R112	191	0R182	1R149
95	0R94	1L113	192	0R149	1R41
96	0R101	0L14	193	0R76	1R158

B The complete prior 194-state frontier

The following inventory contains all 34 old symmetry classes, including auxiliary and rejected machines. Short hashes identify the representative bytes; full hashes, exact paths, and equivalence-class signatures are in `comparison/frontier194.csv`. "Reported" refers to the earlier packet's ordinary mathematical claim and physical evidence, not a new formalization certification by this audit.

Name and short hash	Status	Distinction
CS both-zero6 `28b3a48e3485`	Auxiliary	Known-zero X decrement at both source26 and final emitter exit; research composition.
CSZ-A3 `57a1edbaafb0`	Auxiliary	CSZ plus comparator-return register3 and final-emitter register3 decrements.
CSZ-A5 `2bcebbeefd95`	Auxiliary	CSZ plus comparator-return register3 and final-emitter register5 decrements.
CSZ-A6 `4bccc41f2e84`	Auxiliary	CSZ plus comparator-return register3 and final-emitter register6 decrements.
Cached I-tree zero-jump seed `1dd4bf6a2c6b`	Auxiliary	Alternate decoder plus cached controller and entry zero jump; search basis.
Gamma(D)-CSZA `f9527c1867a0`	Auxiliary	Supplementary Gamma(D) emitter plus known-zero changes.
M4R `a3c1d556d61b`	Reported	Reassociated coefficient4 constructor decoder; original resolved source graph.
M5 `180a0fd1f046`	Reported	Formula constructor coefficient5; explicit variable and proof recoding.
M6 `725ce8474d7b`	Reported	Formula constructor coefficient6; explicit variable and proof recoding.
MPH `f668a2f4de78`	Auxiliary	MP result cached in H while retaining original Shift; physical search input.
PA194 `4bfea62b17aa`	Reported	Separate diagonal head, 67-reader decoder with seven folds, original controller.
PA194-68/8 `9087d70fc0e1`	Reported	Same separate-head source, reassociated 68-reader decoder with eight folds.
PA194-AD `2d65846b3808`	Reported	PA194 with physical A/D register exchange.
PA194A `3c3e74ffcb5d`	Reported	CS entry jump at source26 becomes decrement of known-zero register A.
PA194C `d759313c03b5`	Reported	MP result remains cached in H; Shift tests H=1; original ten-entry logical interface recompiled.
PA194CS `1c71002e5b8a`	Reported	Symmetric Shift guard; cached MP controller; contextual controller fold.
PA194CSZ = PA194X `62e93098caf7`	Reported	CS entry jump at source26 becomes decrement X; two headline names denote identical dense bytes.
PA194CZ `b713e73457be`	Auxiliary	Cached controller; entry known-zero decrement of register 6.
PA194D `4226b7702de5`	Reported	Separate diagonal axiom uses Gamma(D), new proof selector, 69 readers and nine folds.
PA194F `4fcee0c2ef7b`	Reported	CS final emitter jump becomes decrement of known-zero register X.
PA194H `3276a34fd038`	Reported	No-push MP, cached-only Shift; original comparison orientation.3653 source sites,4754 outcomes,57922-step startup.
PA194HS = PA194NS `19935635f2c5`	Reported	No-push MP, cached-only symmetric Shift;3650 source sites,4750 outcomes,57768-step startup. Two submissions deliver identical dense bytes.

Name and short hash	Status	Distinction
PA194J `7c498ced0eb0`	Reported	Ten source jumps become known-zero decrements; three physical rows change.
PA194N `37f7e909724c`	Reported	Successful MP drops dummy-zero push, retains original Shift; revised command proof compiler.
PA194N5 `b521b1086ad2`	Reported	No-push MP with radix5 formula constructors;3715 source sites,4816 outcomes,58320-step startup.
PA194N6 `36da42a81d60`	Reported	No-push MP with radix6 formula constructors;3776 source sites,4877 outcomes,58320-step startup. Two submissions deliver identical dense bytes.
PA194Q `48e98b5504dc`	Reported	Quantifier-distribution curried premises exchanged; K/S proof translation.
PA194R `8a9a8a30773f`	Reported	Reassociated decoder of PA194F; 1111 source paths changed, same resolved source graph.
PA194ZR `11e4b8132038`	Reported	Eight comparator returns and final emitter return become known-zero register3 decrements.
PA194altC `a509952c1e06`	Auxiliary	Cached controller on alternate 68-reader decoder.
Rejected CS node1-X `55ef6a817807`	Rejected	Native103/read1 becomes dec6; exact blank run halts at1361.
Rejected I-tree node1-X `f8abcd947786`	Rejected	Known-zero source rewrite fails blank bootstrap; exact halt at1361.
Rejected ZR node1-X `1c39a0d81ff3`	Rejected	Register6 comparator-return variant; exact blank run halts at1361.
Rejected seven-zero scanner `c48d6711e9c7`	Rejected	Decoder local folds pass, but source-address scanner violates runtime contract.

C Packet-by-packet ledger

The 51 entries cover every filed research packet and every incoming ZIP at the pinned revision. Full source locations are in `comparison/packet_ledger.csv`. Deduplication of headline machine names is handled separately in Appendix B.

No.	Packet	Contribution or status
1	`pa_unary_power_cover`	Unary power covering at PA strength (report of 27 October 2026). The U208 packets carry it as a baseline.
2	`U208_ConPA_Conditional_Context_2026-10-29`	U208 by a conditional tape-context reduction. States a 208-state candidate machine/u208.tm.
3	`U208_ConPA_Entry_Context_2026-10-29`	U208 by eliminating a state using entry context. States a 208-state machine machine/u208.tm.
4	`U208_ConPA_History_2026-10-29`	U208 by a finite-history decoder fold. States a 208-state candidate machine/u208.tm.
5	`U208_ConPA_2026-10-29`	U208 by a three-cell fold. States a 208-state candidate machine/u208.tm.
6	`u208_research_2026-10-29`	U208 research release. States two 208-state machines, machine/u208.tm and machine/variant_plain/u208.tm.
7	`Below_U208_Research_2026-10-29`	Research toward fewer than 208 states. States that no machine with 207 states or fewer was obtained.
8	`u208_comparison_research_2026-10-29`	Comparison of the 208-state proposals and a search below 208. States that the target was not achieved.
9	`ConPA_below208_research_2026-10-29`	Comparative audit and reduction search below 208. States that it does not establish a 207-state machine.

No.	Packet	Contribution or status
10	pa_consistency_research_2026-10-29	Audit and new source reductions. States that no validated machine below 208 states was obtained.
11	pa208_research_2026-10-29	Comparison and research below 208 states. States that it does not establish a valid machine below 208.
12	pa_tm_research_2026-11-13	Comparison and reduction research. States a new 196-state variant with shorter source paths and nothing below 196.
13	pa_consistency_research_2026-11-13	Comparison and diagonal-vacuity research. States a shorter axiom program with a 199-state implementation and nothing below 196.
14	conpa_tm_comparison_and_reduction_search	Comparison and reduction search. States two experimental 196-state command variants and nothing below 196.
15	pa196_research_2026-11-13	PA196 comparison, source factorization and proof obligations. States a source tree of 157 internal nodes and nothing below 196.
16	pa_consistency_research	Research and verification packet. States that no machine below 196 states has been validated.
17	pa_tm_comparison_and_reductions	Comparison and new rewrites, a research snapshot.
18	pa_tm_extended_research_2026-11-13	Comparison and refined reduction search. States that the smallest accepted count is 196.
19	conpa_tm_comparison_and_search	Comparison and a search below 196. States that none was obtained; files named 195 are rejected or diagnostic candidates.
20	pa_consistency_reduction	Comparison and reduction experiments. States two new 196-state variants and a shorter source program with a 203-state implementation.
21	pa_consistency_comparison_2026_11_13	Comparison and new reduction experiments. States that nothing below 196 was obtained.
22	PA_consistency_TM_research_2026-11-13	Comparison and new source reductions. States that no machine below 196 is claimed.
23	pa_consistency_research_2026-11-13_2	Comparison and search below 196. States two new layouts with 196 and 198 states and a command-level reduction.
24	conpa_tm_study_2026-11-13	Comparison and reduction study. States that nothing below 196 was obtained.
25	pa_below196_research	Comparison and reduction experiments. States that nothing below 196 was obtained.
26	pa_consistency_comparison_and_new196_2026-11-13	Comparison and reduction research. States that it contains no accepted machine below 196.
27	pa_tm_research_2026-11-13_2	Comparison and reduction investigation. States same-size work, failed candidates and nothing below 196.
28	pa_consistency_cross_archive	Cross-archive constructions and reduction search. States that it does not certify a 195-state machine.
29	pa195_conpa_2026-11-13	PA195. States a 195-state machine pa195/pa195.tm and an article with a PA proof of the equivalence.
30	pa_consistency_reduction_audit_2026-11-14	Comparison and reduction audit. States nothing below 196 and no new theorem.
31	pa_consistency_research_2026-11-14	Comparison and NEG18. States an 18-instruction encoded negation with a compiled machine of 197 states.
32	PA195_ConPA_research_2026-11-14	A 195-state machine. States the primary machine_195S.tm and a second table machine_195C.tm.
33	pa195_followup_research	Comparison and searches below 195. States a graph-distinct 195-state machine M195S8 and nothing below 195.

No.	Packet	Contribution or status
34	pa194_conpa_2026-11-13	A 194-state machine. States the table construction/tables/pa194.tm and a proof of the equivalence in PA.
35	pa194_conpa_2026-11-13_2	PA194. States a 194-state machine pa194/pa194.tm and an article with a PA proof of the equivalence.
36	pa195_diagonal_combined_2026-11-13	PA195DC. States a different 195-state machine.
37	pa194C_conpa_2026-11-13_2	PA194C. States a materially different 194-state machine, an article with a PA proof of the equivalence, and nothing below 194.
38	pa194_new_designs_2026-11-13	Two new 194-state machines, PA194C and PA194CS. States that nothing below 194 was obtained.
39	pa194c_conpa	PA194C, a cached-head 194-state machine. States that no machine below 194 was established.
40	pa194c_conpa_2026-11-13_1	PA194C. States a different 194-state machine and an article with a PA proof of the equivalence.
41	pa194c_conpa_2026-11-13	PA194C, a cached-head 194-state machine. States an article with a PA proof of the equivalence and nothing below 194.
42	pa194_radix_family_2026-11-13	New 194-state machines calibrated to Con(PA), a radix family.
43	pa194j_pa194d_conpa_2026-11-13	PA194J and PA194D. States two new 194-state machines.
44	pa194q_conpa_2026-11-13	PA194Q and PA194CSZ. States two new 194-state machines and that no machine below 194 was established.
45	PA194R_research_2026-11-14	PA194R. States a different 194-state machine with a new decoder layout and that no machine below 194 was verified.
46	pa194N_conpa_2026-11-13	PA194N. States a new 194-state machine, an article with a PA proof of the equivalence, and nothing below 194.
47	pa194zr_conpa_2026-11-13	PA194ZR. States a new 194-state machine machine/pa194zr.tm, a PA proof of the equivalence, and nothing below 194.
48	pa194hs_conpa_2026-11-13	PA194H and PA194HS: no-push successful MP returns in top; Shift tests cache only. HS uses symmetric comparison; HS is byte-identical to NS from a separate submission. Both194 states, ordinary PA proof and two physical symbolic checkers reported.
49	pa194ns_conpa_2026-11-13	PA194NS is byte-identical to PA194HS; independent submission gives full command compiler and universal implementation checks.194 states; no sub194 result.
50	pa194n5_n6_conpa_2026-11-13	PA194N5 and PA194N6 combine radix5/6 formula constructors with no-push successful MP. Each194 states; standard-library JavaScript universal verifier and ordinary PA proof. N6 duplicates separate N6 research delivery. PDF was typeset directly, not compiled from TeX.
51	pa194n6_research_2026-11-13	PA194N6 duplicates the N6 machine in the N5/N6 packet, with fuller independent physical replay, source-splice review and command/radix proof;194 states, no sub194 result.

References

[1] ProveIt, *PA consistency machines*, pinned README, https://github.com/VladimirReshetnikov/ProveIt/blob/3cc7fbf75ba111a0da696ef2f6001dd528891bd5/Computability/BusyBeaver/Research/PA/README.md.

[2] ProveIt, *Machines in the delivered packets*, pinned `PA/docs/MACHINES.md`; `MACHINES.tsv`, `MACHINE_FILES.tsv`, and `OMITTED.tsv`, same repository revision.

[3] James Chang, imported PA196 source snapshot, `PA/jamesyc-upstream/pa196/`, especially the numerical logic, arithmetic refutation compiler, command compiler and `M6` physical interpreter; source project https://github.com/jamesyc/BusyBeaver.

[4] ProveIt, PA195 and separated-head PA194 research packets in `PA/docs/`; retained proof fragments in `references/logical/original_pa195_calibration.tex`, `diagonal_calculus.tex`, and `separate_head_calculus.tex`.

[5] ProveIt, `PA/docs/pa194_radix_family_2026-11-13/`; retained uniform proof in `references/logical/radix_calibration.tex`.

[6] ProveIt incoming `pa194hs_conpa_2026-11-13.zip` and `pa194ns_conpa_2026-11-13.zip`, pinned revision; no-push/cached-only command compiler and symmetry audit.

[7] ProveIt incoming `pa194n5_n6_conpa_2026-11-13.zip` and `pa194n6_research_2026-11-13.zip`, pinned revision; coefficient-six no-push construction and physical checks.

[8] ProveIt, `Computability/BusyBeaver/Research/ZFC/`, pinned README and the documented K215/K218 context-fold and no-push research.

[9] ProveIt, `Logic/PrimitiveRecursiveArithmetic/ConciseIPRA/`, pinned README, logical interfaces and bi-interpretation program.